School of Computer Science Engineering and Technology

Course- B.Tech

Type- Elective

Course Code- CSET-346

Course Name-Natural Language processing

Year- 2026

Semester- Even

Lab Assignment 1.

Natural Language Processing: Tokenization and Language Modelling

Processing Steps:

- Install the following Python libraries using pip:
  nltk
  spaCy
  textblob
- Load the above libraries in a Python script or Jupyter notebook.
- Write a Python script to tokenize a simple sentence using NLTK.
- Load the English language model in spaCy and Perform POS tagging on the following sentence.
  sentence = "The quick brown fox jumps over the lazy dog."
- Write a Python function to clean text by converting it to lowercase and removing punctuation.

This week's lab assignment is all about implementing your first language model.

You are supposed to download a huge body of text at least (10,000 words). You can download the text from Wikipedia.

- You now must create a model that computes the probabilities of the next word based on unigrams and bigrams.

- Assignment: Given a starting word, you now must predict the whole sentence of length 15 using the ideas of both unigrams and bigrams.